

M45 — Inline device control cards in the encounter Devices card

Phase: Phase 7 follow-on (post-M44, operator feature request) **Status:** **DONE** **Blocked by:** M22 (Devices-card scaffold), M43 (device routes multi-patient aware), M44 (modal scope + cabinet block) **Blocks:** none (room-level cart UI still tracked separately) **Estimated effort:** 0.75 day

1. Purpose

Operator feature request:

"Once the device is added the control system for the device should populate in the device area of the encounter for easy access of the instructor."

Before M45, the encounter console's Devices card was display-only: it polled `/api/room/state` every 2 s and rendered a count summary ("3 device station(s) bound · 0 chat · 1 EHR · 3 device station(s)"). To actually CONTROL a bound device — inject an alarm, clear it, reassign the patient, advance the pump's clock — the instructor had to reopen the M42/M44 *Managed devices* modal every time.

M45 inlines the full control surface. After a device is added (via the modal — that's the right place to mint a QR), it shows up in the Devices card as a card-style row with its own:

- Device kind icon + label + model + online/offline indicator + runtime state
- Patient-assignment dropdown (reassign without opening the modal)
- Active alarm list with per-alarm **Clear** buttons + a **Clear all** button
- Alarm-tone picker + **Inject** button (server-validated tone list)
- For pumps: **+5m** / **+15m** / **+1h** advance-time buttons

The modal stays for adding new devices (mint QR + assign label). All everyday operations now happen in place.

2. Structure

Files touched:

- `portal/static/encounter_console.js`:
- `pollState()` no longer paints the Devices card — it's

refocused on the encounter state badge. - New `pollDevices()` polls `/api/device/roster?join=<jc>` (the M43 multi-patient-aware route). - New `renderDeviceCards(stations)` and `renderDeviceCard(s)` paint the per-device markup. - New action handlers `onDeviceAction(el)` (inject / clear-one / clear-all / advance-time) and `onDeviceAssign(sel)` (reassign). All call the per-station device routes — which since M43 work in multi-patient mode via `station.session_id`. - New `DEVICE_TONE_CATALOG` keyed by `device_kind`. Tone IDs match the server-side `PUMP_ALARMS` / `CABINET_ALERTS` catalogs in `portal/devices/engine/alarms.py` (curated subset of the most common training tones — server validates). - New `KIND_LABEL` for friendly icons + names (■ IV pump, ■ Med cart, ■ Call bell, etc.). - New `personaNameFor(pid)` helper that resolves persona display names from the cached `encVoiceBody.personas` array (built by M33's `bootVoices`). - `startPolling()` now launches a third timer (`devicesTimer`) on its own 3 s cadence. The roster fold is slightly heavier than the 2 s state/transcript

poll (each station does an engine.fold + alarm scan), so we space it out.

- portal/static/encounter_console.css:
- Appended .device-card block style (overrides the older

```
.device-list li { display: flex } that would have wrapped the card contents horizontally). -
.device-card-header / .device-card-meta / .dev-dot (online/offline indicator) /
.device-card-assignment / .dev-alarm-list / .dev-tone-picker / .dev-btn-sm /
.dev-btn-inject / .dev-time-group.
```

No backend change. No new HTML template markup. The encounter console's existing `<ul id="device-list">` (M22 scaffold) is the container; M45 just renders richer markup into it.

3. Uses

3.1 Operator flow

[illegible]

3.2 Polling cadence

Loop	Cadence	What it fetches
Telemetry (M22/M23)	1 s	GET /api/encounter/{id}/telemetry
State + transcript (M22/M30)	2 s	/api/room/state + /api/encounter/{id}/transcript
Devices (M45)	3 s	/api/device/roster?join=<jc>

The polls share the `visibilitychange` pause hook — when the console tab is hidden, all three timers stop.

4. Functions (exported API surface)

Symbol	Where	Purpose
pollDevices() (new)	portal/static/encounter_console.js	Fetches /api/device/roster?join=<jc>, hands to renderer.

Symbol	Where	Purpose
<code>renderDeviceCards(stations)</code> (new)	same	Maps each station to a card LI.
<code>renderDeviceCard(s)</code> (new)	same	Per-station card markup.
<code>onDeviceAction(el)</code> (new)	same	Handles inject / clear-one / clear-all / advance from the per-card buttons.
<code>onDeviceAssign(sel)</code> (new)	same	Patient-reassign dropdown handler.
<code>DEVICE_TONE_CATALOG</code> (new)	same	Per-kind tone list — must match server <code>alarms.py</code> catalog.
<code>KIND_LABEL</code> (new)	same	Per-kind friendly label with emoji.

5. Limitations

- **Cabinet (med cart) reassignment is not exposed inline.** The

card shows a `""Med cart (room-level)""` note under the assignment dropdown. The full shared-cart UI ships in the deferred M44-§8 work (room-level med carts + linking + grouped MAR).

- **Hard-coded tone catalog.** If the server adds new tones in

`alarms.py` they won't appear in the dropdown until the JS catalog is updated. Server validates every inject so a stale client gets a clear 400. A future M46 could fetch the catalog from a new `/api/device/tones` route.

- **No WS push.** The card refreshes on the 3 s poll. WS push

for sub-second updates exists at `/ws/instructor` (M16-era); wiring it would replace the polling loop. Out of scope; the M22 polling cadence is the bedside-station latency budget anyway.

- **Persona dropdown reuses `encVoiceBody.personas` populated**

by `bootVoices()`. If `bootVoices()` failed (e.g. server returned 4xx), the dropdown only shows `"— unassigned —"`. That matches the v6 behavior; the operator can still reassign by id via the Managed-devices modal.

- **advance_time buttons are pump-only.** Cabinets and future-

device buttons don't have a clock to advance. The buttons are conditionally rendered.

- **Polling is per-tab.** Two operators on different machines

each poll independently. Acceptable; same as M22.

6. Test status

Test file	Asserts	Status	Last run
<code>tests/v7/test_inline_device_cards.py::test_encounter_js_has_pollDevices_function</code>	<code>pollDevices, /api/device/roster?join=, renderDeviceCards</code> in source	PASS	2026-05-27
<code>...::test_encounter_js_renders_per_device_card_with_controls</code>	All 5 action data-attrs (inject, clear-one, clear-all, advance, assign) present	PASS	2026-05-27

Test file	Asserts	Status	Last run
...::test_encounter_js_wires_actions_to_device_routes	onDeviceAction calls /inject, /clear, /advance_time; onDeviceAssign calls /assign	PASS	2026-05-27
...::test_encounter_js_tone_catalog_covers_supported_kinds	Catalog covers pump_iv / pump_enteral / cabinet with real tone IDs (occlusion_downstream, low_battery, discrepancy_alert)	PASS	2026-05-27
...::test_encounter_js_polls_devices_on_its_own_cadence	startPolling launches pollDevices with DEVICES_POLL_MS; stopPolling clears devicesTimer	PASS	2026-05-27
...::test_encounter_js_does_not_call_assignment_dropdown_on_cabinet_inline	Cabinet rows include the room-level note	PASS	2026-05-27
...::test_device_roster_returns_full_card_shape	/api/device/roster?join= returns station_id, device_kind, device_model, label, online, character_id, active_alarms, runtime_state	PASS	2026-05-27
...::test_inject_endpoint_returns_ok_in_multi_patient	POST /api/device/<sid>/inject returns 200+ok=true with a real tone ID	PASS	2026-05-27
...::test_encounter_console_devices_card_has_list_anchor	Devices card still has #device-list UL + #button-manage-devices button	PASS	2026-05-27
Full v7 suite	310 passed, 1 skipped (M20 Playwright skip — unchanged)	PASS	2026-05-27

7. Change list

Date	Author	Change	Files
2026-05-27	claude-code	Initial M45 implementation: pollDevices, renderDeviceCards, renderDeviceCard, onDeviceAction, onDeviceAssign; DEVICE_TONE_CATALOG + KIND_LABEL; 3 s poll cadence; device-card CSS; 9 new tests	portal/static/encounter_console.js, portal/static/encounter_console.css, tests/v7/test_inline_device_cards.py (new)

8. Open questions / known issues

- **WS push instead of poll.** /ws/instructor already broadcasts

every device event; wiring it to the inline cards would give sub-second alarm/clear/assign updates. A future M46 could swap the polling loop for a WS subscriber (with poll as fallback when WS disconnects).

- **Tone-catalog drift.** Defer the *fetch catalog from server*

refactor until either the catalog grows past ~12 tones per kind OR a tone gets renamed server-side. Today the curated subset covers the high-frequency training alarms.

- **Shared med carts + grouped MAR** still tracked separately as

the M44-S8 deferred work. M45 unblocks everyday device operations; the shared-cart UI is independent and queued for a future module.

- **Advance-time when no channel is running** returns 200 from

the server but the pump's `running_channels` list stays empty. The card doesn't surface that "applied = 0 minutes" feedback yet — operator would need to look at the v6 device-detail modal in the Managed-devices iframe. Low priority.

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